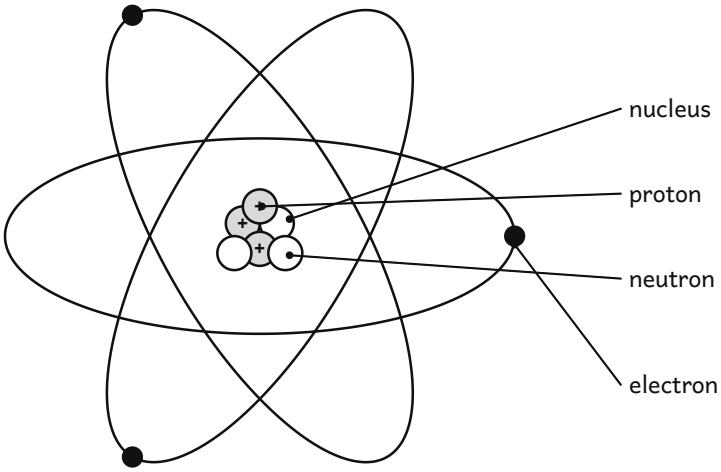
PHYSICAL AND CHEMICAL SCIENCE

Atoms and Their Structure

Everything is made of atoms: protons, neutrons, electrons, their mass and charge, and how we found out



In this pack

Learn it: Atoms: the building blocks of everything ★☆☆
Fact check: Parts of an atom ★☆☆
Think it through: Thinking about atoms ★★☆
Investigate: Build and compare atom models ★★☆
Real-world case study: How did we find out about atoms? ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · C3 · ATOMS

Atoms: the building blocks of everything

Name: Class: Date:

Learning goals

- Explain that all matter is made of atoms.
- Describe and model the parts of an atom: nucleus, protons, neutrons and electrons.
- Compare the mass and charge of protons, neutrons and electrons.

Imagine cutting a piece of gold in half, then in half again, and again, and again. Could you go on for ever? About 2400 years ago, a Greek thinker called **Democritus** said no: in the end you would reach a tiny piece that cannot be cut. He called it *atomos*, which means "uncuttable". Today we know he was nearly right.

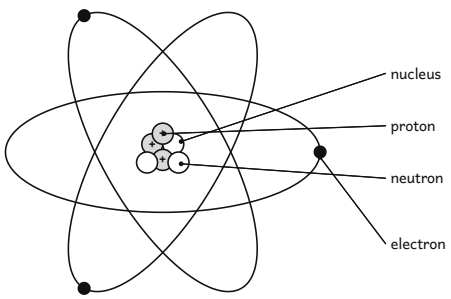
Everything is made of atoms

Your body, the air, the Sun, a mango, a mobile phone: all matter is made of **atoms**. Atoms are incredibly small. The full stop at the end of this sentence contains millions and millions of them. There are about **118 kinds** of atom, one for each **element**.

Inside an atom

Atoms are made of even smaller particles:

- The **nucleus** is the tiny centre of the atom. It holds **protons** and **neutrons**.
- **Protons** have a **positive** (+) electric charge.
- **Neutrons** have **no** charge (they are neutral).
- **Electrons** have a **negative** (−) charge. They move very fast in the space around the nucleus.



A model of an atom (not to scale).

Comparing the particles

Particle	Where it is	Charge	Mass
proton	in the nucleus	positive (+)	heavy (1 unit)
neutron	in the nucleus	none (0)	heavy (1 unit)
electron	around the nucleus	negative (−)	very light (about 2000 times lighter than a proton)

An atom has the **same number** of protons and electrons, so the + and − charges balance and the atom has no overall charge.

Did you know?

An atom is mostly **empty space**! If the nucleus were the size of a marble in the middle of a football field, the electrons would be like tiny specks of dust flying around the edges of the stadium.

Different atoms, different elements

The **number of protons** decides which element an atom is. Every hydrogen atom has 1 proton; every carbon atom has 6; every oxygen atom has 8; every gold atom has 79.

Element	Protons	Neutrons (usual)	Electrons
hydrogen	1	0	1
helium	2	2	2
carbon	6	6	6
oxygen	8	8	8

Atoms join together

Atoms can join to make **molecules**. Two hydrogen atoms and one oxygen atom join to make one molecule of **water** (H_2O).

Models help us think

Nobody can see inside an atom with their eyes. Scientists use **models**: drawings, balls and sticks, or computer pictures. Models help us understand, but they are not exactly like the real thing. Real electrons do not move in neat circles, and the nucleus is far smaller than in our drawings.

Try this at home

Make a model atom of carbon with clay or paper: 6 protons and 6 neutrons stuck together in the middle, and 6 electrons around them.

Key words

atom	the tiny building block of all matter
nucleus	the centre of an atom
proton	a particle in the nucleus with a positive charge
neutron	a particle in the nucleus with no charge
electron	a very light particle with a negative charge, around the nucleus
charge	positive (+) or negative (−) electricity
molecule	two or more atoms joined together
model	a way of showing something we cannot see

FACT CHECK · C3 · ATOMS

Parts of an atom

★☆☆
Grade 4

Name: Class: Date:

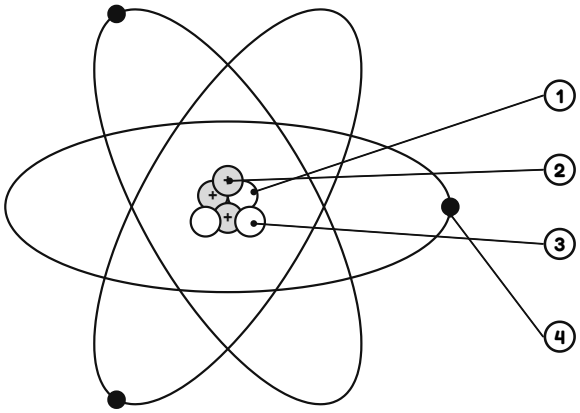
Learning goals

- Label an atom.
- Recall charges.

1 Label the atom. [4]

LABELS

neutron nucleus electron proton



1 _____ 2 _____

3 _____ 4 _____

2 Complete the table. [6]

Particle	Charge	Where?
proton		
neutron		
electron		

3 Fill in the gaps. [4]

WORD BANK

protons neutron nucleus electron atoms cells

All matter is made of _____. The centre of an atom is the _____. The number of _____ decides which element an atom is. The lightest particle is the _____.

4 **True or false?** [4]

Statement	True	False	If false, correct it
1. Neutrons have no charge.			
2. Electrons are heavier than protons.			
3. There are about 118 kinds of atom.			
4. An atom is mostly empty space.			

THINK IT THROUGH · C3 · ATOMS

Thinking about atoms

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Use the atom model.
- Compare particles.

1 **An atom of nitrogen has 7 protons. How many electrons does it have? Explain.** [2]

2 **Most of the mass of an atom is in the nucleus. Explain why.** [2]

3 **Two atoms both have 6 protons. Are they the same element? Explain.** [2]

4 **Give one way the drawing of an atom on the Learn it page is useful and one way it is not like a real atom.** [2]

INVESTIGATE · C3 · ATOMS

Build and compare atom models

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Make models from data.
- Find patterns in data.
- Evaluate models.

You will need

modelling clay or paper circles in three colours (one for protons, one for neutrons, one for electrons), card, glue.

Make models of these atoms. Use the data to count the particles.

1 **Complete the table.** [8]

Element	Protons	Neutrons	Electrons	Total particles	Mass (units)
helium	2	2			
lithium	3	4			
carbon	6	6			
oxygen	8	8			

2 **What pattern do you notice between the number of protons and the number of electrons?** [1]

3 **Why do we not count the electrons when we work out the mass?** [1]

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LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C3 · ATOMS

How did we find out about atoms?

★★★
Grade 4

Name: **Class:** **Date:**

Learning goals

- Learn how ideas change with new evidence.
- Order events on a timeline.
- Think about creativity in science.

Science · History · Maths

Nobody has ever seen inside an atom with their eyes. So how do we know what atoms are like? Over hundreds of years, scientists did clever experiments and changed their models when they found new **evidence**.

In **1803**, **John Dalton** in England said that each element is made of its own kind of atom, like tiny hard balls. In **1897**, **J. J. Thomson** discovered the **electron**, so atoms could not be solid balls. In **1911**, **Ernest Rutherford** fired tiny particles at a thin sheet of gold. Most went straight through, but a few bounced back! This showed that atoms are mostly empty space with a tiny, heavy **nucleus**. In **1932**, **James Chadwick** discovered the **neutron**.

Based on the history of atomic models.

1 **Put the discoveries in order on the timeline.** [4]

- ☐ Chadwick discovers the neutron (1932)
- ☐ Democritus names the atom (about 400 BCE)
- ☐ Rutherford discovers the nucleus (1911)
- ☐ Dalton: elements are made of atoms (1803)
- ☐ Thomson discovers the electron (1897)

2 **Maths. How many years passed between Dalton's idea and Chadwick's discovery?** [1]

.....

3 **Rutherford was surprised when some particles bounced back. Why did this make him change the model?** [2]

4 **Scientists combine evidence with creative thinking. Explain what this means using the story of the atom.** [2]

Show what you know

★★★
Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

F	B	B	R	D	A	T	O	M	J
E	P	E	M	I	F	L	P	A	J
L	R	L	O	S	O	C	N	W	G
E	O	E	D	U	C	H	E	H	L
C	T	M	E	A	D	A	U	M	O
T	O	E	L	H	M	R	T	G	M
R	N	N	T	O	A	G	R	Y	N
O	W	T	K	H	S	E	O	D	R
N	D	B	G	S	S	W	N	A	B
S	M	N	U	C	L	E	U	S	W

atom charge electron element mass model neutron nucleus proton

2 Which particle has a negative charge? [1]

- ◯ proton
- ◯ neutron
- ◯ electron
- ◯ nucleus

3 Which two particles are found in the nucleus? [1]

- ◯ protons and electrons
- ◯ protons and neutrons
- ◯ neutrons and electrons
- ◯ only electrons

4 Draw a model of a helium atom (2 protons, 2 neutrons, 2 electrons). Label each part. [3]

Helium atom

5 **Challenge. Explain why an atom has no overall charge even though it contains charged particles.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain that all matter is made of atoms	○	○	○
label the parts of an atom	○	○	○
compare the charge and mass of the particles	○	○	○
make atom models from data	○	○	○
explain how models of the atom changed	○	○	○
say why models are useful but not perfect	○	○	○